Principles and Anti-Patterns of the Monte Carlo Simulations

Principles

1. Always be clear what probability distribution (PD) you have used for the Monte Carlo Simulation and justify its selection as the selection of this has a large bearing on the accuracy of the simulation output. In addition, it is recommended that the PD is selected based on empirical evidence.

2. Graphs are very useful as they show how quickly the numbers approach the expected value, plus any outliers, etc.

3. The quality of the pseudo-random number generator can have a large effect on the simulation output – so it is worth knowing how it is generated. (Burmaster and Anderson, 1994).

Anti-patterns

1. Consider carefully the tools used to run the simulation. As mentioned above, the Monte Carlo method (MCM) can be very sensitive to the pseudo-random number algorithm used, and pre-packaged add-ins or tools give no control over what is used.

2. The sampling method can have a big impact, which again can be related back to point 1.

3. MCM does not work well with non-standard distributions, or indeed if the user mismatches the PD and the real-world distribution of data.

4. Because it is computationally intensive, many-dimensional simulations do not work well. (Marsden, 2018)